Module_1:

Team Members:

Megala Kanna

Project Title:

Neurodegenerative - Longer project

We want to explore how the brain physiology and environment can influence protein buildup in the brain. We will compare data on brain pH to the samples of amyloid beta and tau proteinopathies. We want to observe if there is a statistically significant correlation, p < .05, between these markers.

Project Goal

This project seeks to see if there is a statistically significant correlation between the brain's pH and the concentration of amyloid beta & tau tangles?

the questions we considered were:

how does years of education correlate with age of onset cognitive symptoms?

How does brain pH correlate to age of onset cognitive symptoms?

How does Tau protein correlate to age of death?

Is there a tipping point of concentration of amyloid tangles/ tangles to the age of diagnosis (heterogeneity quality of detection method)?

Do females show better compensation phenotypes have a higher amyloid tangles/ tangles concentration? (might be able to do because we available data on if they have children or not)

Disease Background:

Fill in information about 5 studies.

- Prevalence & incidence: 12 million Americans have Alzheimer's disease (AD), approximately 1.25 million 75-85y and 170,000 y. prevalence
  https://www.alzheimerfamilies.com/alzheimer-s-statistics-how-many-people-have-alzheimer-s#:~:text=%2D%20Approximately%2012%20million%20Americans,-Incidence&text=%2D%20Approximately%2012%20million%20Americans%20have,%2D%20Approximately%2170%2C000%20new%20cases. incidence
  deathcharts.com
  journals.lww.com/alzheimerjournal/fulltext/2020/01000/Trends__Incidence__and_Prevalence_of_Alzheimer%27s_disease_among.7.aspx#:~:text=Alzheimer%27s%20disease%20%28AD%29,%2C%20and%20Hispanic%20Americans%2C%20%2C%20%2C%20